KU LEUVEN

Master of Engineering: Architecture
Master of Civil Engineering
Master of Mechanical Engineering

H0T88A Structural Optimization

Mattias Schevenels

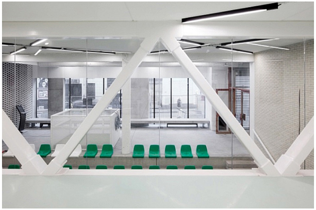
KU Leuven
Faculty of Engineering Science
Department of Architecture

https://architectuur.kuleuven.be/structopt

General introduction

Structural design optimization

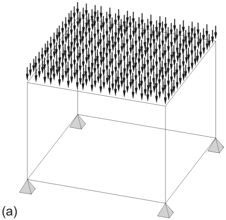
Aim: minimize weight / cost / environmental impact of structures, taking into account structural performance and buildability criteria.

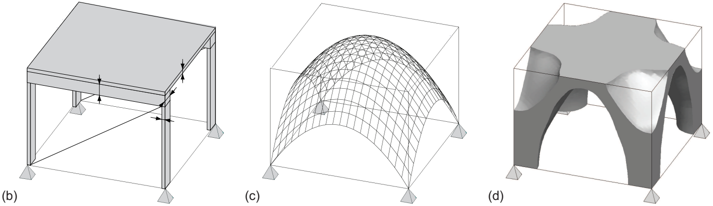


KU LEUVEN

Design optimization strategies

Design domain





Size optimization

Shape optimization

Topology optimization

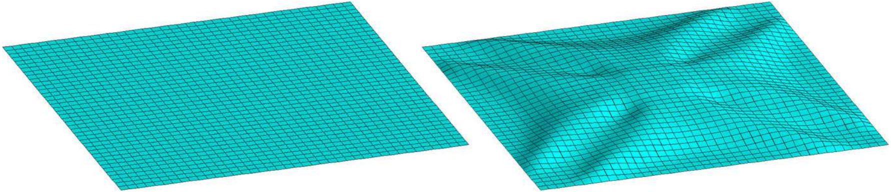
KU LEUVEN

Size optimization

– The designer fixes the layout of the structure.
– Member dimensions are optimized.

Shape optimization

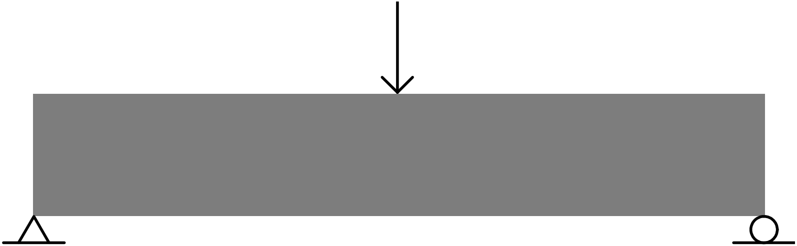
– The structural shape is described by a (large) number of parameters.
– These parameters are optimized.



K.-U. Bletzinger, M. Firl, J. Linhard, and R. Wüchner.
Optimal shapes of mechanically motivated surfaces.
Computer Methods in Applied Mechanics and Engineering, 199(5-8):324-333, 2010.

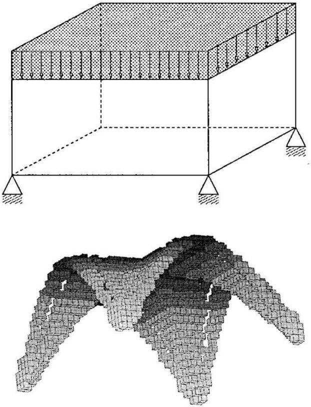
Topology optimization

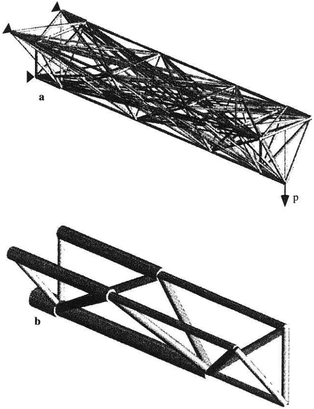
– The design domain, loads, boundary conditions, and amount of material are specified.
– The optimal material distribution in the design domain is determined.



Topology optimization

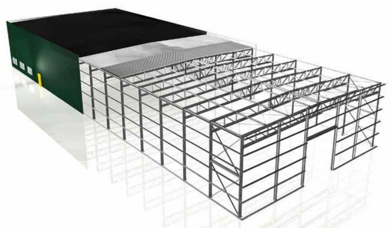
- Continuum topology optimization.
- Truss topology design.





Optimization of steel structures using standard sections

Size, shape and topology optimization of real structures following the Eurocodes.
Discrete or mixed discrete-continuous design optimization.



R. Van Mellaert. Optimal design of steel structures according to the Eurocodes using mixed-integer linear programming methods. PhD thesis, Department of Architecture, KU Leuven, 2017.

W. Dillen. Eurocode-compliant optimal design of steel structures. PhD thesis, Department of Architecture, KU Leuven, 2020.

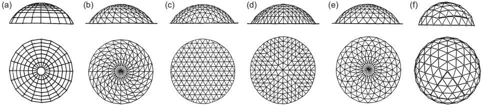
Optimization of strained gridshells

Shape optimization of a simply supported timber gridshell (result: 4 times stiffer than a similar gridshell designed by classical formfinding techniques).

J. Rombouts, G. Lombaert, L. De Laet, and M. Schevenels. A novel shape optimization approach for strained gridshells: Design and construction of a simply supported gridshell. Engineering Structures, 192:166-180, 2019.

Optimization of unstrained gridshells

Size, shape and topology optimization of reticulated domes and shells.

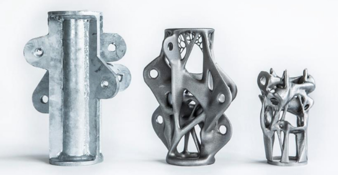


W. Gythiel, C. Mommeyer, T. Raymaekers, and M. Schevenels. A comparative study of the structural performance of different types of reticulated dome subjected to distributed loads. Frontiers in Built Environment, 6:56, 2020.

KU LEUVEN

Topology optimization of 3D-printed parts

Topology optimization of a 3D-printed joint for a tensegrity structure, taking into account minimum length scale and maximum overhang angle constraints.

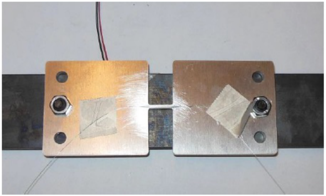


S. Galjaard, S. Hofman, N. Perry, and S. Ren. Optimizing structural building elements in metal by using additive manufacturing. In Proceedings of the International Association for Shell and Spatial Structures (IASS), 2015.

J. Pellens, G. Lombaert, B. Lazarov, and M. Schevenels. Combined length scale and overhang angle control in minimum compliance topology optimization for additive manufacturing. Structural and Multidisciplinary Optimization, 59(6):2005-2022, 2019.

High-accuracy strain sensors

Topology optimization of displacement transducers to improve the accuracy of strain gauges for structural health monitoring (result: strain upscaling factor of 73).

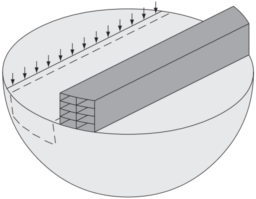


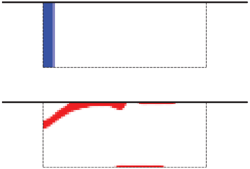
T. Jonckers. Development of high-accuracy strain sensors using topology optimization. Master's thesis, Department of Civil Engineering, KU Leuven, 2016.

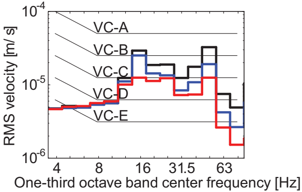
KU LEUVEN

Vibration isolating screens in the soil

Topology optimization of a 2D wave barrier to reduce railway traffic induced vibrations in buildings.



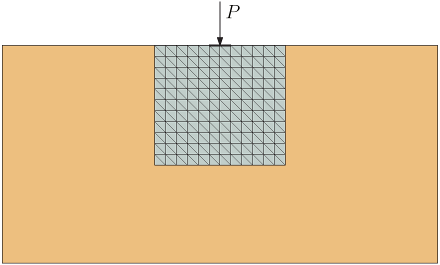


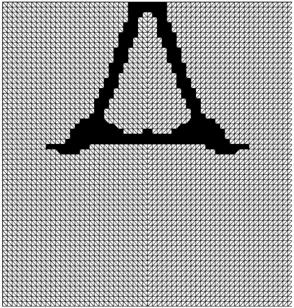


C. Van hoorickx, O. Sigmund, M. Schevenels, B.S. Lazarov, and G. Lombaert. Topology optimization of 2D wave barriers. Journal of Sound and Vibration, 376:95-111, 2016.

Geotechnical structures

Topology optimization of a jet-grouted foundation accounting for the ultimate limit state.





S. François, W. Stalmans, and M. Schevenels. Topology optimization of geotechnical designs involving ultimate limit states. Proceedings of the 6th European Conference on Computational Mechanics and the 7th European Conference on Computational Fluid Dynamics, Glasgow, United Kingdom, June 2018.

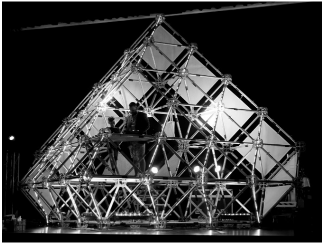
Optimization of reinforced-concrete ribbed floors

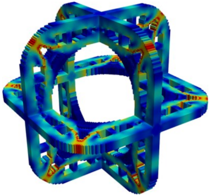
Performance-based and Eurocode-based optimization of concrete floors similar to Nervi's floor in the Gatti wool factory.

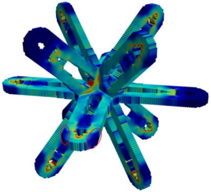
T. Barbier, M. Schevenels (supervisor), and G. Lombaert (cosupervisor). Performance-based and Eurocode-based design optimization of reinforced concrete ribbed floors. PhD thesis, Department of Architecture, KU Leuven, 2023.

A modular building system for temporary structures

Topology optimization of a joint for modular space frames constructed from standard scaffolding bars.



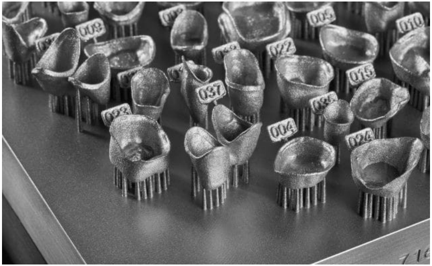


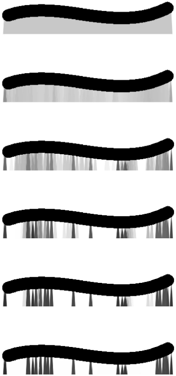


D. Van Cauteren, M. Schevenels (supervisor), G. Lombaert (cosupervisor), and K. Allacker (cosupervisor). A new modular building system for temporary structures consisting of reusable components. PhD thesis, Department of Architecture, KU Leuven, 2023.

Optimization of support structures for 3D-printed parts

Topology optimization of cone supports to counteract thermal deformations in selective laser melting (in collaboration with Materialise).





D. Deferm. Topology optimization with feature mapping for the design of cone support structures for thermal displacements in SLM. Master's thesis, Department of Architecture, KU Leuven, 2022.

Gradient-based optimization algorithms

A variety of algorithms is available to solve the optimization problem:

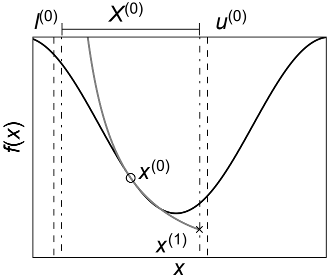
First-order methods

- Optimality Criteria method (OC).
- Steepest descent method (with penalization).
- CONvex LINearized method (CONLIN).
- Method of Moving Asymptotes (MMA).
- ...

Second-order methods

- Newton method (with penalization).
- Sequential Quadratic Programming (SQP).
- ...

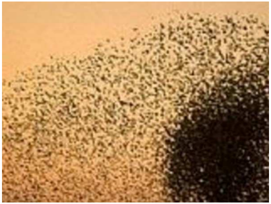
OC and MMA are the most common methods in topology optimization.



Metaheuristic optimization algorithms

Metaheuristic algorithms are optimization methods inspired by some physical or biological phenomenon. They are based on random trial and error.






KU LEUVEN

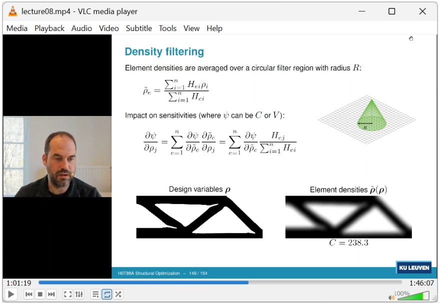
Course objectives

- To be able to appreciate the need for numerical optimization to improve the efficiency of a structural design.
- To be able to mathematically formulate a structural design optimization problem.
- To have an understanding of the most useful design optimization techniques.
- To be able to use these techniques for practical size, shape, and topology optimization problems.

Course format

- Blended learning format.
- One introductory on-campus session (today).
- The remainder of the course is organized in learning modules.
- Core material is provided through web lectures.

The course is structured around clearly defined module checkpoints (every 1–2 weeks).



Structure of a learning module

Each module consists of:

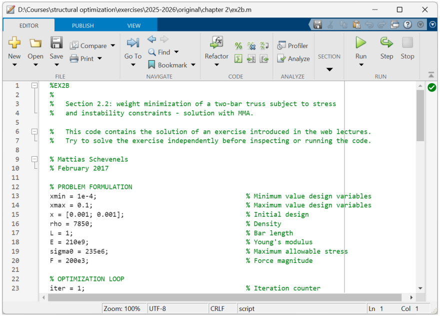
- 1–2 web lectures (theory + MATLAB examples and exercises).
- A short homework task (self-check).
- One on-campus guidance session.

Modules build on each other. After each guidance session, the course moves on.

How you are expected to work

- Study the web lectures independently and actively.
- Pause during MATLAB exercises and solve them yourself.
- Use slides, textbook and optional technical notes where needed.
- Complete the homework as a self-check.

Guidance sessions are for questions and feedback, not for repeating the web lectures or providing worked solutions.



Attendance and responsibilities

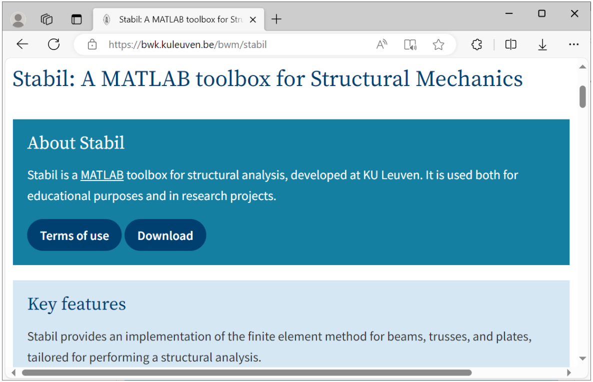
- Attendance at guidance sessions is not mandatory.
- If you do not attend, you are expected to ensure independently that you have reached the required level.
- Guidance sessions are not recorded.

Attendance is mandatory for:

- Assignment hand-out session (at least one group member).
- Final presentation session.

Prerequisites

- We use the Stabil toolbox for MATLAB for finite element analysis; Toledo provides access to the Stabil website where the toolbox can be downloaded.
- It is expected that those who have not previously worked with Stabil become familiar with it on their own (read the first 9 pages of the manual and do some MATLAB experiments).
- Prior knowledge of the finite element method and experience with MATLAB are required.



Evaluation

Assignment solved in groups of 3 students using MATLAB:

- Formulation of an optimization problem.
- MATLAB implementation.
- Critical interpretation of the results.
- Written report + oral presentation.

Exact dates and practical details are available on Toledo.

After the announcement of the results, it is possible to make an appointment (by email) for feedback.

Course material

All course resources are available via Toledo:

- Learning module descriptions.
- Web lectures (core material).
- Slides.
- Textbook (Christensen & Klarbring) and papers.
- MATLAB toolboxes (StructOpt, Stabil).
- MATLAB exercises and examples.
- Optional technical notes.